EXERCISE 1

[3 marks]

What is the last digit of 7^{217} ?

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EXERCISE 2

[5 marks]

Expand and simplify :

1. $(5^x - 2)(5^x - 4)$ **2.** $(x^{\frac{1}{2}} + 2)(x^{\frac{1}{2}} - 2)$ **3.** $(5 - 2^{-x})^2$

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EXERCISE 3

1. Factorise $4^x + 9(2^x) + 18$ [3 marks]

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2. Simplify $\frac{5^{n+1} - 5^n}{4}$ [2 marks]

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EXERCISE 4 Solve for x :

1. $3 \cdot 2^{x+1} = 24$ [4 marks]

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2. $12 \cdot 3^{-x} = \frac{4}{3}$ [4 marks]

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3. $4 \cdot \left(\frac{1}{3}\right)^{x} = 36$ [4 marks]

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4. $25^{x} - 23(5^{x}) - 50 = 0$ [4 marks]

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EXERCISE 5 Consider the two exponential functions $f(x) = 555(1.2)^{x} + 3$ and $g(x) = 555(1.1)^{x}$.

1. Which function is steeper ? Explain. [2 marks]

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2. For each function, state its horizontal asymptote. [2 marks]

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3. Determine whether the function is increasing or decreasing, and explain. [4 marks]

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EXERCISE 6 [5 marks]

A species of bear is introduced to a large island off Alaska where previously there were no bears. 6 pairs of bears were introduced in 1998. It is expected that the population will increase according to $B_t = B_0 \times 2^{0.18t}$ where t is the time since the introduction.

1. Find B_0.

2. Find the expected bear population in 2018.

3. Find the expected percentage increase from 2008 to 2018.

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Answer Key

1. Last digit of 7^{217}

The last digits of powers of 7 repeat every 4 and $217 = 4 \times 54 + 1$ so $\boxed{7}$

2. Expand and simplify

a. $(5^x - 2)(5^x - 4) = (5^x)^2 - 6 \cdot 5^x + 8 = \boxed{5^{2x} - 6 \cdot 5^x + 8}$.

b. $(x^{1/2} + 2)(x^{1/2} - 2) = (x^{1/2})^2 - 4 = \boxed{x - 4}$.

c. $(5 - 2^{-x})^2 = 25 - 10 \cdot 2^{-x} + (2^{-x})^2 = \boxed{25 - 10 \cdot 2^{-x} + 2^{-2x}}$.

3. **a.** Factorise $4^x + 9(2^x) + 18$

Let $X = 2^x$, then $4^x = (2^x)^2 = X^2$: $X^2 + 9X + 18 = (X + 3)(X + 6)$. so $\boxed{(2^x + 3)(2^x + 6)}$

b. Simplify $\dfrac{5^{n+1} - 5^n}{4}$ $\frac{5^n(5-1)}{4} = \frac{4 \cdot 5^n}{4} = \boxed{5^n}$.

4. Solve for x

a. $3 \cdot 2^{x+1} = 24 \therefore\ 2^{x+1} = 8 = 2^3 \ \therefore\ x + 1 = 3 \Rightarrow \boxed{x = 2}$.

b. $12 \cdot 3^{-x} = \dfrac{4}{3} \therefore\ 3^{-x} = \frac{4}{36} = \frac{1}{9} = 3^{-2} \Rightarrow \boxed{x = 2}$.

c. $4\left(\dfrac{1}{3}\right)^x = 36 \ \therefore\ \left(\frac{1}{3}\right)^x = 9 = 3^2 = 3^{-(-2)} \Rightarrow \boxed{x = -2}$.

d. $25^x - 23(5^x) - 50 = 0$ Let $X = 5^x$, then $25^x = (5^2)^x = 5^{2x} = X^2$: $X^2 - 23X - 50 = 0 = (X - 25)(X + 2)$.

Since $X = 5^x > 0$, we keep $X = 25$: $5^x = 25 = 5^2 \Rightarrow \boxed{x = 2}$.

5. Exponential functions

$f(x) = 555(1.2)^x + 3, \qquad g(x) = 555(1.1)^x$

a. Steeper : $\boxed{f}$ since $1.2 > 1.1$.

b. Horizontal asymptotes : $\boxed{y = 3 \text{ for } f}$ and $\boxed{y = 0 \text{ for } g}$.

c. Both are increasing because their bases are greater than 1.

6. Bear population model

$B_t = B_0 \cdot 2^{0.18t}$

a. 6 pairs were introduced $\Rightarrow B_0 = \boxed{12}$.

b. Year 2018 : $t = 2018 - 1998 = 20 \Rightarrow B_{20} = 12 \cdot 2^{0.18 \cdot 20} = 12 \cdot 2^{3.6} \approx \boxed{145}$.

c. Percentage increase from 2008 to 2018 :

2008 : $t = 10$, 2018 : $t = 20$

$\frac{B_{20} - B_{10}}{B_{10}} \times 100 = \left(\frac{2^{3.6}}{2^{1.8}} - 1\right) \times 100 = \left(2^{1.8} - 1\right) \times 100 \approx \boxed{249\%}$.